Lecture 5.5: VS Code, GitHub, and the Data Section

AREC 570 / ECON 530 · Unit 5

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Today’s sequence

First third: VS Code as the research workspace

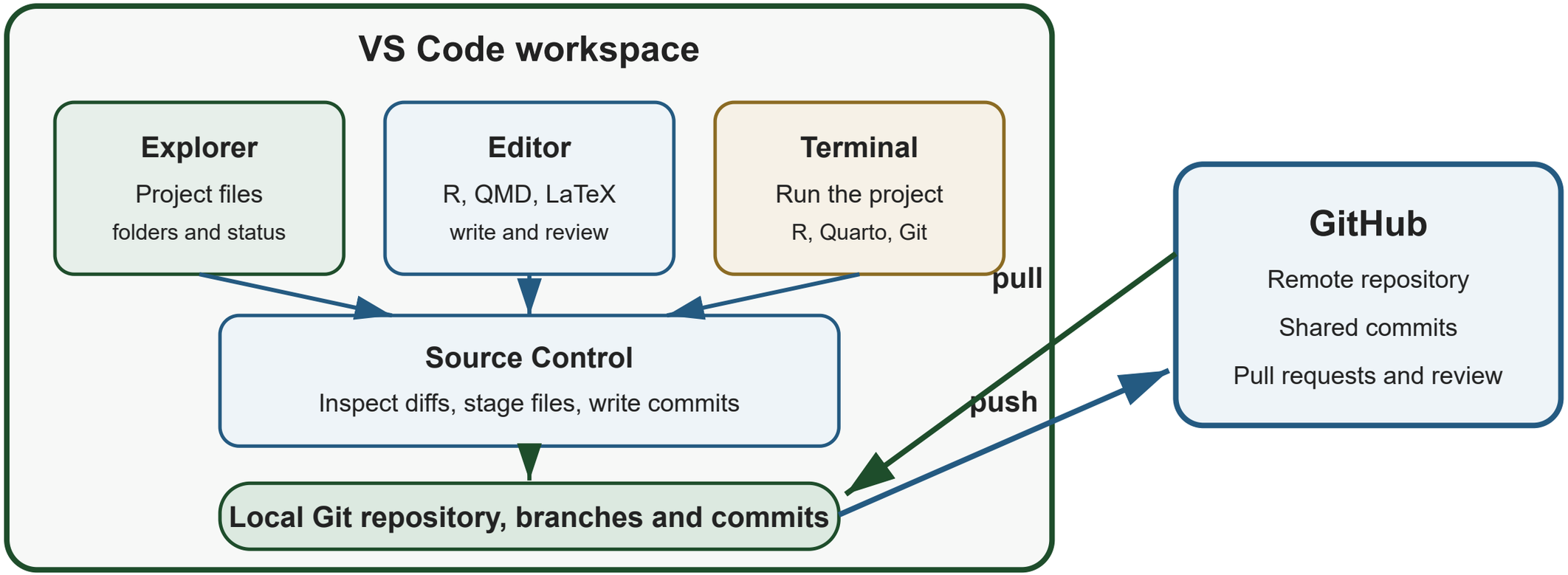
- Open and run a research repository
- Review changes and use Git
- Connect local commits to GitHub

Remaining two thirds: the data section and its repository evidence

- Establish data fit for the hypotheses
- Document the sample and variable construction
- Develop the empirical setting with descriptive statistics
- Link manuscript claims to code and outputs

VS Code as a research workspace

The IDE brings the workflow into one window



Required tools for the VS Code workflow

Tool	Role in the workflow	Check
Git	Local version history	`git --version`
R	Statistical computing	`R --version`
VS Code	Editor and integrated workspace	Open the application
R extension	R terminal, completion, plots, and help	Create an R terminal
Quarto extension	Edit, execute, render, and preview `.qmd` files	Preview a `.qmd` file

Install the R language server from an R console

```
1 install.packages("languageserver")
```

Opening the repository root

1. Clone the repository or locate its existing local folder
2. Select **File > Open Folder** and choose the repository root
3. Review the active branch in the lower-left status bar
4. Open the integrated terminal and run `git status`

The repository root contains `.git` and the project files

Workspace checks before editing

VS Code area	What to verify
Explorer	Project structure and expected files
Status bar	Active branch and repository status
Source Control	Uncommitted and staged changes
Terminal	Working directory and available software
Extensions	R and Quarto support enabled for this workspace

Start the session by pulling accepted commits into the current branch

R execution within VS Code

Open the Command Palette with `Ctrl+Shift+P`

1. Run **R: Create R terminal**
2. Open an `.R` file
3. Select code and press `Ctrl+Enter`

Run the master script from the R terminal

```
source("code/run_all.R")
```

Paths and the project root

- Use paths relative to the repository root
- Place computer-specific settings in an ignored configuration file
- Record package requirements and external software
- Provide one documented command that rebuilds the analysis
- Direct generated outputs to predictable folders

A collaborator should be able to identify the entry point from the README

The Git cycle in Source Control

1. Pull the current branch and switch to a task branch
2. Edit and run the project
3. Inspect each changed file and stage the intended changes
4. Commit with a message that describes the research change
5. Push the branch to GitHub

The Source Control view and terminal operate on the same local Git repository

Source: Source control in VS Code

Diff review before committing

Ask four questions

- Does every changed file belong to this task
- Do the additions implement the intended research decision
- Did an output, credential, or large data file enter the change accidentally
- Can the commit message explain the change without opening every file

Stage a coherent set of files after reviewing the diff

GitHub features in VS Code

Built-in Git support handles cloning, pulling, committing, and pushing

The **GitHub Pull Requests and Issues** extension adds

- Pull-request creation
- File-by-file review
- Inline comments
- Local checkout of a proposed change

Repository permissions still determine where a branch can be pushed

Source: Work with GitHub in VS Code

VS Code Demo

1. Open the repository and locate the analysis entry point
2. Identify the active branch and remote
3. Open one R script and one LaTeX file
4. Edit a demonstration branch, inspect the diff, and commit
5. Push the branch and locate it on GitHub

Writing the data section

Research question and testable hypotheses

Research question

> How did grocery retailers adjust prices and dollar margins during COVID-19 states of emergency, when anti-price-gouging laws, demand, and wholesale costs changed at the same time?

Framework-derived hypotheses

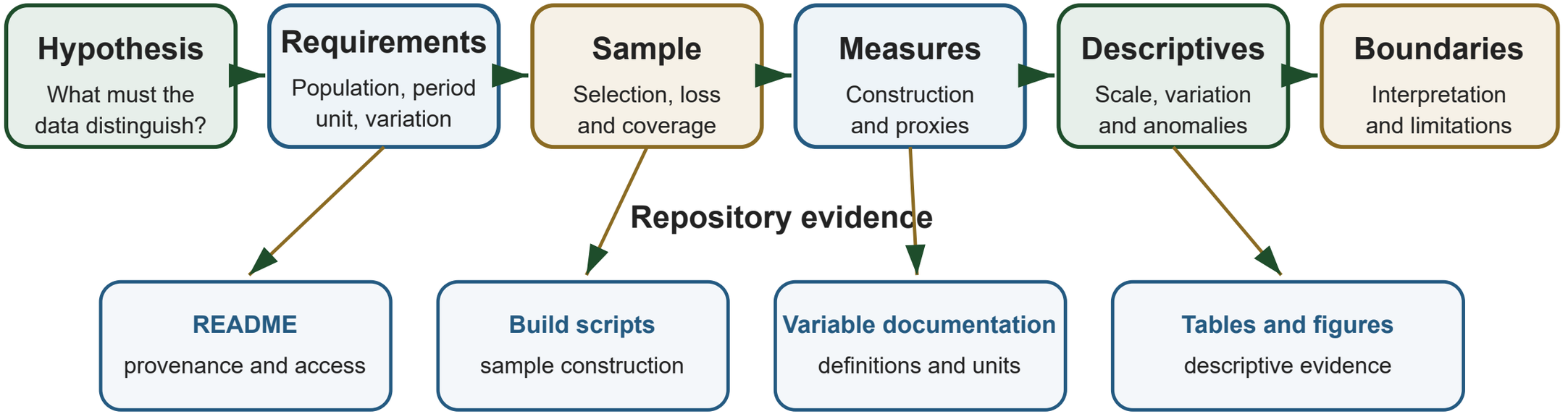
1. Retail prices and dollar margins differ across the pre-SOE, SOE, and post-SOE periods
2. Within-chain price differences vary with state enforcement status if APG protections affect pricing
3. Retail prices respond to wholesale acquisition costs, and pass-through may change during the SOE
4. More frequent promotions can reduce net transaction prices while posted prices remain similar

The data section makes an empirical argument

The section should establish that the observed data can evaluate the hypotheses under stated conditions

- The population and setting correspond to the research question
- The sample contains the observations and variation required by the hypotheses
- The measures represent the concepts in the framework
- The construction rules can be reproduced
- The limitations define the scope of interpretation

Evidence chain for a data section



Hypotheses determine data requirements

Part of a hypothesis	What the data section must establish
Outcome	The measure, unit, and reference period
Explanatory factor	The observed variable that represents the proposed mechanism
Comparison	The groups, places, or periods that generate variation
Population	The units to which the hypothesis applies
Conditions	The controls, restrictions, and assumptions needed for interpretation

The hypothesis determines the observations, measures, and variation the data section must establish

A complete source description

Address the questions that determine provenance and coverage

- Who collected or assembled the data
- Why the data were collected
- When and where collection occurred
- Who or what was observed
- How observations were recorded
- Which population the source covers
- Which access, licensing, or confidentiality conditions apply

Add information that affects interpretation or replication

Selection rules define the target of inference

Report

- The population from which the sample is drawn
- Every inclusion and exclusion rule
- The order in which the rules are applied
- The observations lost at each consequential step
- Whether attrition or missingness differs across groups or time
- How the analytic sample compares with the population of interest

Explain how each rule affects the claims the sample can support

Define the observation unit and key

The observation unit states what one row represents

The key is the field or combination of fields that uniquely identifies each row

- Connect the observation unit to the actors, outcomes, and timing in the hypotheses
- Explain any aggregation from the source records to the analytic unit
- State how records from different sources are matched
- Verify that the key is unique and nonmissing

Variable construction connects concepts to fields

For each central variable report

- The concept from the framework
- The source fields
- The unit and reference period
- The aggregation or transformation
- The interpretation of the resulting value
- The treatment of missing or implausible values
- The limitations of the measure

Give enough information for another researcher to reconstruct the variable

Missing data and sample loss

For every consequential restriction state

- Which field is missing or invalid
- How many observations are affected
- Whether the loss is concentrated by place, time, product, or group
- How the analysis treats the affected observations
- How the restriction changes the interpretation of the sample

Imputation documentation

Report

- Which variables contain imputed values
- The unit and time window used for imputation
- The algorithm and predictors
- The assumptions required by the method
- The number and share of values imputed
- An indicator that identifies imputed observations
- A sensitivity analysis using observed values

Keep the original value and the imputed value in separate fields

Proxies and empirical scope

A proxy is an observed measure used to represent an unobserved concept

For each proxy

- Name the concept the hypothesis requires
- Define the observed measure
- Explain why the measure represents the concept
- Identify the relevant dimensions the measure omits
- Limit the empirical claim to what the measure can support

Measurement limits should appear in the data section and in the interpretation of the results

Example: price-gouging study

Data provenance and scope

The opening paragraph identifies the source, records, period, locations, firms, and geography

> We use item-level scanner data from DecaData, which report daily retail transactions from January 1, 2018, through July 17, 2023 for 549 grocery stores affiliated with three major retail chains operating across five Southeastern U.S. states

What the opening sentence establishes

- The data provider
- The type and frequency of the records
- The observation period
- The number and type of locations
- The number of chains
- The geographic coverage

The remaining source description explains the chains, product records, and access conditions

Sample entry and selection rules

The price-gouging manuscript reports a three-stage product-selection procedure

1. Restrict the universe to 47 perishable fruit and vegetable categories
2. Apply a coverage screen around policy activation and expiration
3. Rank passing products by sales and volume and select five items

Coverage screen: present in 80 percent of both 26-week windows, observed for five weeks on each side, and passing in 75 percent of stores

Analytic sample construction

Stage	Result
Item-level transactions	74 product categories
Perishable produce restriction	47 candidate categories
Coverage screen	Products observed near activation and expiration
Sales and volume ranking	Five focal products
Analytic restrictions	Positive quantity, price, and wholesale cost
Final panel	604,121 store-product-week observations

The manuscript should report the counts available at each consequential stage

Observation unit and key

The analysis aggregates transactions to the **store-product-week** level

Index	Meaning
i	Store
j	Product
s	State
t	Week

The key is the combination of store, product, and week

State identifies the policy environment associated with each store

Net retail price construction

For store i, product j, state s, and week t

$$p_{jist}^{net} = \frac{\sum_{d \in t} net_sales_{ijsd}}{\sum_{d \in t} volume_{ijsd}}$$

- The numerator is total transaction revenue after promotional discounts
- The denominator is total units or pounds sold
- The ratio is the average price paid at the register
- Units follow the product, pounds for random-weight produce and units for packaged lettuce

Cost and margin construction

Average wholesale acquisition cost

$$w_{jist} = \frac{\sum_{d \in t} total_cost_{ijsd}}{\sum_{d \in t} volume_{ijsd}}$$

Retail-wholesale margin

$$M_{jist}^{net} = p_{jist}^{net} - w_{jist}$$

The shared product, store, week, and physical unit give the spread a direct interpretation

Policy measures

The empirical setting determines three periods for each state

- **Pre-SOE** weeks before the emergency declaration
- **SOE** weeks when the declaration and associated APG protections were active
- **Post-SOE** weeks after the declaration expired

SOE_{st} links the state-level legal setting to each store-product-week

Dur_{st} counts weeks since activation during the emergency period

Report the date source and the rule used to assign partial weeks

Sample restrictions and missing observations

The analysis requires positive quantities, retail prices, and wholesale costs

Price-change models also require consecutive observations for the same store and product

The data section should report

- The number of observations removed by each restriction
- The concentration of losses by product, store, state, and period
- The implications for the analytic sample and each hypothesis

Proxies, interpretation, and scope

Construct	Available measure	Scope of interpretation
Consumer price	Net transaction price	Average register price for the item within the week
Retailer input cost	Wholesale acquisition cost	Procurement cost recorded in the point-of-sale system
Retail margin	Net price minus acquisition cost	Retail-wholesale spread
Policy exposure	Active state emergency declaration	Statutory APG exposure by state and week

Labor, logistics, handling, shrink, and spoilage remain outside the observed cost measure

Limitations and empirical claims

The manuscript identifies three central boundaries

- Wholesale acquisition cost captures one component of retailer cost
- Category-level averages can change with the composition of products purchased
- Policy exposure varies at the state-week level across five states

These boundaries guide the language used for margins, price indexes, and policy interpretation

Price and margin claims

Empirical claim	Data feature that supports it	Boundary
Prices changed during the SOE	Weekly net price for the same products and stores	Other contemporaneous changes remain part of the reduced-form setting
Margins changed during the SOE	Retail price and acquisition cost use the same unit	The spread omits other retailer costs

Uniformity and pass-through claims

Empirical claim	Data feature that supports it	Boundary
Pricing uniformity changed	Multiple stores within the same chain and week	Store coverage changes over time
Pass-through changed	Retail and wholesale price changes are observed together	Wholesale costs may reflect shared upstream conditions

Descriptive statistics as evidence

Descriptive evidence required for the hypotheses

- The scale and composition of the analytic sample
- The variation required to evaluate the hypotheses
- The distribution and units of the central measures
- Differences across periods or comparison groups
- Extreme values, missingness, and panel imbalance
- Patterns that motivate later estimation

Design the table after deciding which empirical questions it should answer

A descriptive paragraph has four jobs

1. **Orient** the reader to the sample, unit, period, and comparison
2. **Identify** the pattern that matters for the research question
3. **Quantify** its magnitude with a small number of interpretable values
4. **Qualify** the pattern with sample composition or measurement information

Roles of tables, figures, text, and the repository

Element	Job
Table body	Complete values, units, and sample sizes
Table notes	Definitions, weights, transformations, and sample restrictions
Figure	Time patterns, distributions, or comparisons
Main text	Interpretation tied to the research question
Repository	Code that constructs every reported value

Check that statistics compared in the text use the same sample and unit

Example: descriptive evidence in the price-gouging study

Establish the scale and composition

The descriptive section begins with the analytic scale

> The panel covers 604,121 observations across 549 unique store locations and three retail chains operating in five Southeastern states from January 2018 through July 2023

It then reports changes in store coverage over time and identifies which retailers account for the decline

This information prepares the reader to evaluate panel balance and sample composition

Describe patterns relevant to the hypotheses

The manuscript connects three pieces of descriptive evidence

> Upon the SOE activation in March 2020, total weekly volume rose sharply, increasing 27.4 percent relative to the four preceding weeks

> Both nominal and real prices fell immediately at activation and remained below their pre-SOE range for the duration of enforcement

> Wholesale costs were generally unchanged across periods for most products. Dollar margins narrowed during the SOE for all five products

How the descriptive discussion is organized

Job	Evidence used in the example
Orient	Sample size, stores, chains, states, and period
Identify	Volume, price, cost, and margin patterns during the SOE
Quantify	A 27.4 percent increase in volume and product-level changes
Qualify	Changes in store coverage and differences across products

The discussion gives the reader context for the regression results that follow

The repository supports the data section

The paper and repository answer connected questions

Manuscript question	Repository evidence
Where do the data come from	README data overview and access documentation
How is the sample constructed	SQL build and panel-construction scripts
How are variables defined	README variable table and transformation code
How are exclusions applied	Analysis-sample filters in code
Where do descriptive values come from	Ordered scripts that write tables and figures

The README is the entry point

The price-gouging repository documents

- The research question and empirical setting
- Required SQL builds and local configuration
- The command that runs the analysis
- Data coverage and access restrictions
- Core variables and their interpretation
- Known limitations
- The mapping from scripts to manuscript sections and outputs

A new collaborator should know where to begin after reading the README

Scripts mirror the data argument

Repository artifact	Data-section role
`code/00_read_in_data.R`	Retrieves source tables and assembles the initial panel
`code/02_build_panel.R`	Constructs measures, timing variables, and the estimation sample
`code/03_descriptive_tables.R`	Produces coverage and descriptive tables
`code/run_all.R`	Runs the scripts in the intended order
`code/sql/`	Documents upstream construction of the source tables
`tables_latex/` and `figures/`	Store manuscript-ready outputs

Git tracking by access and purpose

Track in Git	Keep outside Git
Source code and SQL	Licensed transaction records
README and variable documentation	Credentials and connection strings
Package and software requirements	Computer-specific configuration
Small metadata and public inputs	Large intermediate data files
Selected tables and figures	Regenerable bulk outputs

Use `.gitignore` for restricted, sensitive, computer-specific, and regenerable files

Document how an authorized user supplies each required local input

Computing environment

Record

- R and external software requirements
- Package names and versions when version differences matter
- Database or API dependencies
- Required environment variables or configuration fields
- The order of scripts
- Random seeds for stochastic procedures
- The expected output folders

Provide a safe example configuration with placeholder values when local settings are required

Manuscript statements and repository artifacts

Manuscript statement	Supporting artifact
Final sample contains 604,121 observations	Summary table produced by `03_descriptive_tables.R`
The key is store-product-week	Panel-construction code and README data structure
Net price is transaction revenue divided by volume	Variable-construction code and README definition
Five products pass the selection procedure	Product coverage output and selection code
Descriptive means differ across SOE periods	Period-means table and figure scripts

This mapping helps a coauthor, reviewer, or future researcher verify a statement efficiently

Repository audit

- Can I identify the project purpose and empirical setting
- Can I find the analytic unit, key, sample period, and central variables
- Can I distinguish source data, intermediate data, and final outputs
- Can I locate every sample restriction and transformation
- Can I reproduce the descriptive tables from documented inputs
- Can I identify restricted materials and the process for authorized access
- Can I connect each manuscript table and figure to the script that produces it

Apply the workflow to your project

Fit-to-hypothesis matrix

Choose one hypothesis from your conceptual framework

Component	Your project
Hypothesis	What relationship will be evaluated
Required observations	Who or what must be observed
Required variation	Across whom, where, or when
Constructed measures	How the concepts become variables
Sample restrictions	Which observations enter the analysis
Main limitation	Which interpretation requires care
Repository evidence	Which file documents or implements the step

Data-section spine

Write one or two sentences for each heading

1. **Source and provenance**
2. **Coverage and population**
3. **Sample selection**
4. **Observation unit and key**
5. **Variable construction**
6. **Missing data and imputation**
7. **Limitations and scope**
8. **Descriptive evidence**

Group workshop

Choose one volunteer project and work through the evidence chain

1. State one hypothesis
2. Identify the population and required variation
3. Define the observation unit and sample boundaries
4. Explain one central variable construction
5. Identify one proxy or measurement limitation
6. Name the repository artifact that should support each step

Preparation for the next project meeting

Prepare two items for your own project

- A five-sentence outline covering source, sample, observation unit, central measures, and main limitation
- A repository checklist naming the README content, build script, variable documentation, and first descriptive output still needed

Bring both items to the next project meeting or advising conversation

Summary

VS Code brings the project files, R or Quarto execution, Git, and GitHub into one workspace

The data section establishes provenance, sample construction, variable construction, and interpretive scope

Descriptive statistics develop the empirical setting and quantify the variation relevant to the hypotheses

The repository provides the documentation and code needed to verify and reproduce the data section

Resources and course navigation

- Price-gouging research repository
- Source control in VS Code
- R in VS Code
- Quarto in VS Code
- All lectures
- Schedule and preparation
- Course reading list